EE 4122

VLSI Technology Lab

Experiment 01

Schematic Design and Implementation of Logic Gates and Logic Equations Using DSCH2 Software

COURSE TEACHERS

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OVERVIEW

01	**Schematic Design and Implementation of Logic Gates and Logic Equations Using DSCH2 Software**	THIS LAB
02	**Design and Implementation of CMOS Logic Circuit Mask Layouts** *Using Microwind Software*	
03	**Implementation of Logic Gates and Boolean Logic Equations** *Using Hardware Description Language in ModelSim*	
04	**Design and Implementation of Sequential Circuits** *Using Verilog in ModelSim*	
05	**Implementation of Logic Operations Using VHDL** *For FPGA Device in Quartus II*	

ASSESSMENT

10%	Attendance
40%	Weekly Test in the classroom
10%	Report Writing
20%	Quiz
20%	Central Viva

EXPERIMENT 01

Objective & Theoretical Background

Objective

- Design basic CMOS logic gates (NOT, NAND, NOR) at the transistor level.
- Simulate gate behaviour and verify truth tables using DSCH2 software.
- Realize compound Boolean logic equations using combinations of gates.
- Analyze the input/output waveform response of each designed circuit.

What is DSCH2?

DSCH2 is a schematic-capture and logic simulator used to design digital circuits at the transistor and gate level, verify functionality with built-in logic simulation, and prepare designs for physical layout in Microwind.

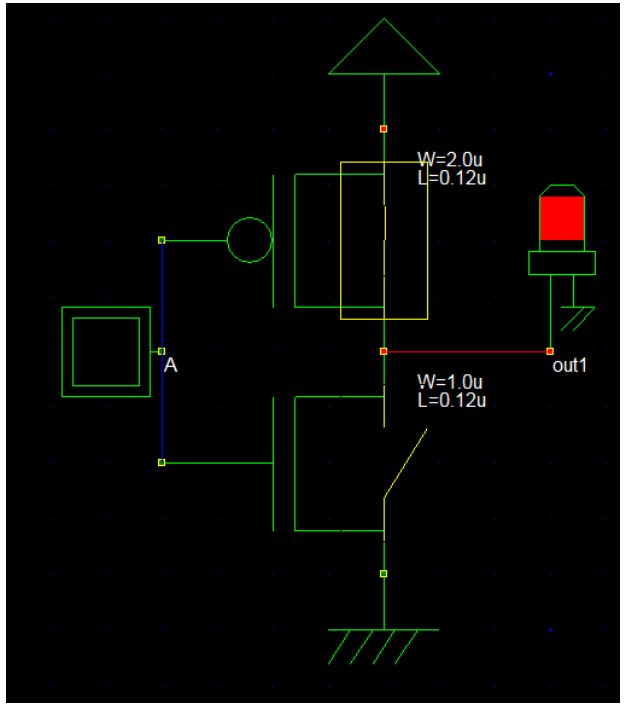
CMOS Logic Design Flow

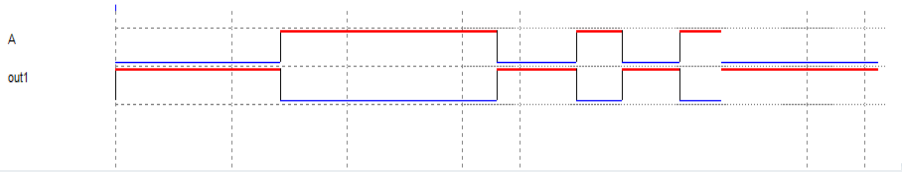
- Boolean equation
- Transistor-level schematic
- Logic simulation
- Waveform verification

BASIC LOGIC GATE

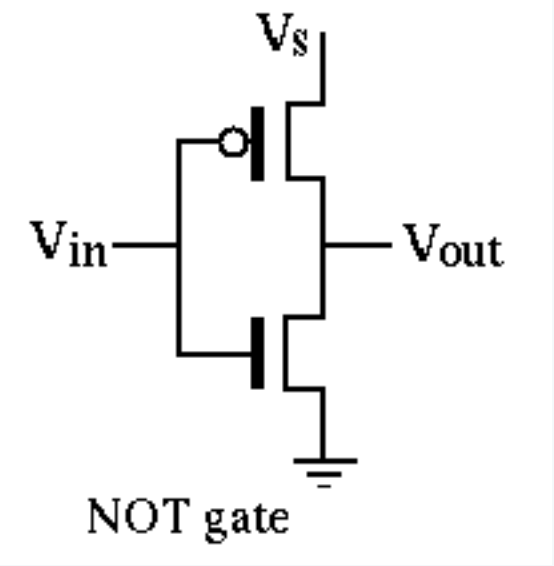
NOT Gate — CMOS Inverter

Transistor-Level Schematic (DSCH2)





Logic Symbol



Truth Table (Y = A')

A	Y
0	1
1	0

The CMOS inverter uses one PMOS (pull-up) and one NMOS (pull-down) transistor. When A = 0, the PMOS conducts and pulls Y to logic 1; when A = 1, the NMOS conducts and pulls Y to logic 0.

TWO-INPUT LOGIC GATES

NAND & NOR Gates — Truth Tables

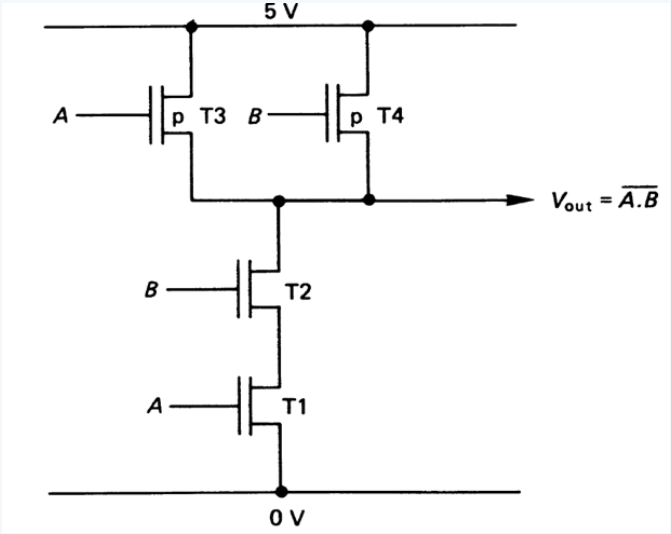
NAND Gate (Y = (A·B)')

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

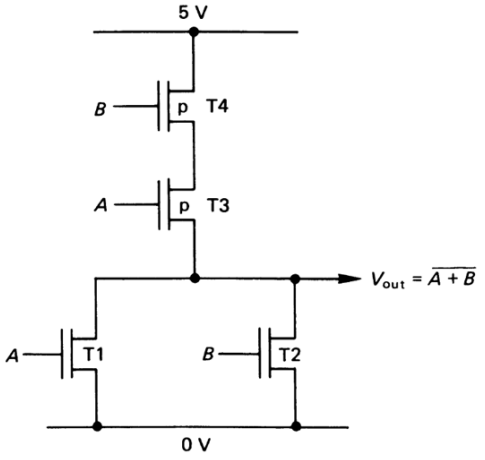
NOR Gate (Y = (A+B)')

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Simulated Waveform — NAND

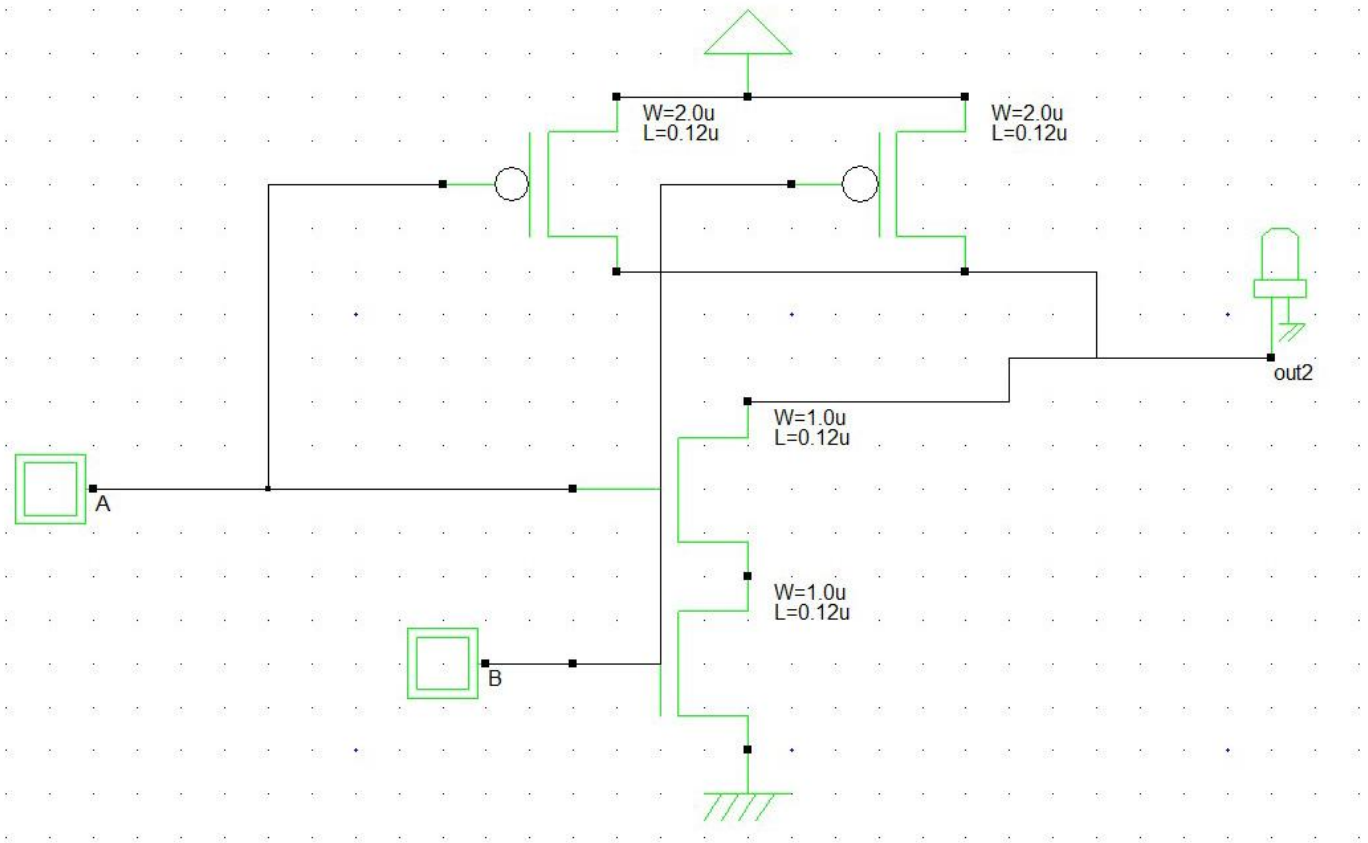


Simulated Waveform — NOR



DSCH2 SCHEMATIC CAPTURE

NAND Gate — Transistor-Level Layout

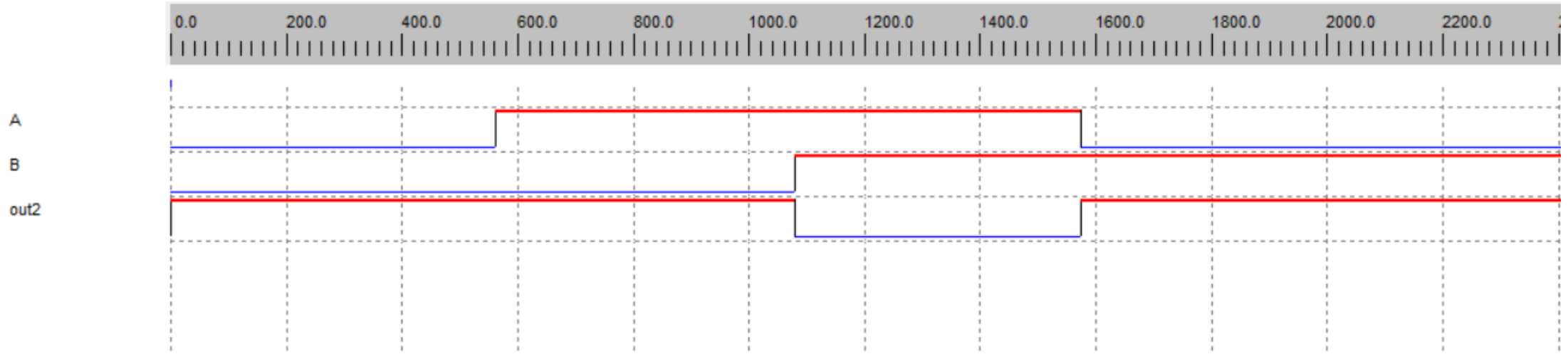


Design Notes

- Two PMOS transistors in parallel form the pull-up network.
- Two NMOS transistors in series form the pull-down network.
- Output is high unless both inputs A and B are logic 1.
- Layout built and verified directly in the DSCH2 schematic editor.

WAVEFORM VERIFICATION

Simulation Result — NAND Output



The waveform confirms correct NAND behaviour: the output stays high for every input combination except when both A and B are simultaneously logic 1, where it briefly falls low — matching the truth table on the previous slide.

5-VARIABLE LOGIC EQUATION

$$Y = (A + B \cdot C + D \cdot E)'$$

A	B	C	D	E	Y
0	0	0	0	0	1
0	0	0	0	1	1
0	0	0	1	0	1
0	0	0	1	1	0
0	0	1	0	0	1
0	0	1	0	1	1
0	0	1	1	0	1
0	0	1	1	1	0
0	1	0	0	0	1
0	1	0	0	1	1
0	1	0	1	0	1
0	1	0	1	1	0
0	1	1	0	0	0
0	1	1	0	1	0
0	1	1	1	0	0
0	1	1	1	1	0

A	B	C	D	E	Y
1	0	0	0	0	0
1	0	0	0	1	0
1	0	0	1	0	0
1	0	0	1	1	0
1	0	1	0	0	0
1	0	1	0	1	0
1	0	1	1	0	0
1	0	1	1	1	0
1	1	0	0	0	0
1	1	0	0	1	0
1	1	0	1	0	0
1	1	0	1	1	0
1	1	1	0	0	0
1	1	1	0	1	0
1	1	1	1	0	0
1	1	1	1	1	0

Reading the Equation

Y = A + B·C + D·E is an OR of three terms: input A alone, the AND of B and C, and the AND of D and E. Y is logic 1 whenever any one of these three terms is true.

Gate-Level Realization

- AND gate → B·C
- AND gate → D·E
- 3-input OR gate combines A, B·C, D·E

4-VARIABLE LOGIC EQUATION

$$Y = (A \cdot B + C \cdot D)'$$

A	B	C	D	Y
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0

A	B	C	D	Y
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Reading the Equation

Y = (A·B + C·D)' is the complement of an OR-of-ANDs expression — a NAND applied to the outputs of two AND terms, A·B and C·D. Y is logic 0 only when at least one AND term is true.

Gate-Level Realization

- AND gate → A·B
- AND gate → C·D
- NOR gate combines the two AND outputs